PROBLEMS **Week: 16 April**

All the indicated numbers refer to the "Problems" section in Serway; e.g. P1.9 is problem 9 from chapter 1.
Alle nommers verwys na die "Problems" afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

The last section, heading, "FROM CHAPTER 13: For week of 23 April 2018", will be done during the tutorial session of the following week.

=======================CHAPTER 7=======================

OQ7.1; OQ7.2 ; OQ7.3; OQ7.4; OQ7.6; OQ7.7; OQ7.10; OQ7.12; OQ7.16; CQ7.1;

CQ7.2; CQ7.3; CQ7.7; CQ7.10; CQ7.11; CQ7.14;

P7.11 (a) 16.0 J (b) 36.9°
P7.13 7.05 m, 28.4°

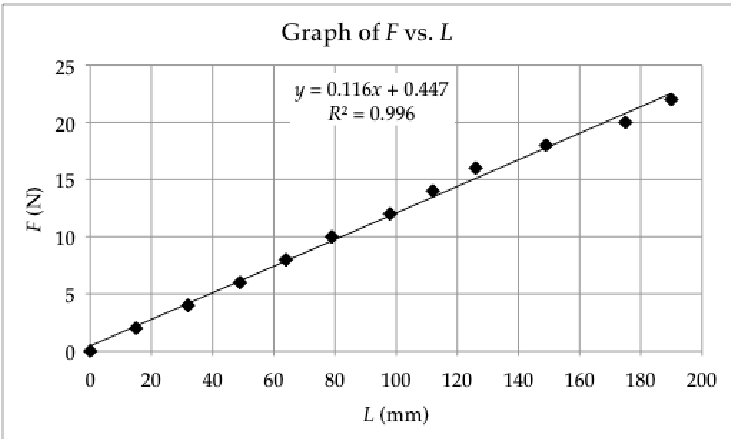
P7.14 (a) 24.0 J (b) -3.00 J (c) 21.0 J

P7.15 (a) 7.50 J (b) 15.0 J (c) 7.50 J (d) 30.0 J

P7.19 (a) 575 N/m (b) 46.0 J

P7.26 (a) Graph (b) -12.0 J

P7.27 (a) Graph



ANS FIG. P7.27(a)

(b) By least-squares fitting, its slope is 0.116 N/mm = $\boxed{116 \text{ N/m}}$.

(c) To draw the straight line we use all the points listed and also the origin. If the coils of the spring touched each other, a bend or nonlinearity could show up at the bottom end of the graph. If the spring were stretched "too far," a nonlinearity could show up at the top end. But there is no visible evidence for a bend in the graph near either end.

(d) In the equation $F = kx$, the spring constant k is the slope of the

F-versus-x graph.

$$\boxed{k = 116\ \text{N/m}}$$

(e) $F = kx = (116\ \text{N/m})(0.105\ \text{m}) = \boxed{12.2\ \text{N}}$

P7.28 (a) 9.00 kJ (b) 11.7 kJ (c) The work is greater by 29.6%.

P7.37 (a) 4.56 kJ (b) 4.56 kJ (c) 6.34 kN (d) 4.22 km/s^2

(e) 6.34 kN (f) The forces are the same. The two theories agree.

P7.41 (a) +2.5 J (b) -9.8 J (c) -12 J

P7.43 (a) -196 J (b) -196 J (c) -196 J (d) The results should all be the same, since gravitational force is conservative

P7.44 (a) $\vec{\mathbf{F}} \cdot (\vec{\mathbf{r}}_f - \vec{\mathbf{r}}_i)$, which depends only on the end points, and not on the path

(b) 35.0 J

P7.51 (a) 40.0 J (b) -40.0 J (c) 62.5 J

P7.58 $W = k_1 \dfrac{x_{\max}^2}{2} + k_2 \dfrac{x_{\max}^3}{3}$

P7.61 (a) F1 = $(20.5\hat{\mathbf{i}} + 14.3\hat{\mathbf{j}})$ N F2 = $(-36.4\hat{\mathbf{i}} + 21.0\hat{\mathbf{j}})$ N

(b) $(-15.9\hat{\mathbf{i}} + 35.3\hat{\mathbf{j}})$ N (c) $(-3.18\hat{\mathbf{i}} + 7.07\hat{\mathbf{j}})$ m/s^2 (d) $(-5.54\hat{\mathbf{i}} + 23.7\hat{\mathbf{j}})$ m/s

(e) $(-2.30\hat{\mathbf{i}} + 39.3\hat{\mathbf{j}})$ m (f) 1.48 kJ (g) 1.48 kJ

(h) The work-kinetic energy theorem is consistent with Newton's second law, used in deriving it.

P7.62 (a) $b = 1.80$ $a = 4.01 \times 10^4\ \text{N/m}^{1.8}$

(b) 295 J

P7.63 The component of the weight force parallel to the incline, $mg \sin\theta$, accelerates the block down the incline through a distance d until it encounters the spring, after which the spring force, pushing up the incline, opposes the weight force and slows the block through a distance x until the block eventually is brought to a momentary stop. The weight force does positive work on the block as it slides down the incline through total distance $(d + x)$, and the spring force does negative work on the block as it slides through distance x. The normal force does no work. Applying the work-energy theorem,

$$K_i + W_g + W_s = K_f$$

$$\frac{1}{2}mv_i^2 + mg\sin\theta(d+x) + \left(\frac{1}{2}kx_{\text{sp},i}^2 - \frac{1}{2}kx_{\text{sp},f}^2\right) = \frac{1}{2}mv_f^2$$

$$\frac{1}{2}mv^2 + mg\sin\theta(d+x) + \left(0 - \frac{1}{2}kx^2\right) = 0$$

Dividing by m, we have

$$\frac{1}{2}v^2 + g\sin\theta(d+x) - \frac{k}{2m}x^2 = 0 \quad \rightarrow$$

$$\frac{k}{2m}x^2 - (g\sin\theta)x - \left[\frac{v^2}{2} + (g\sin\theta)d\right] = 0$$

Solving for x, we have

$$x = \frac{g\sin\theta \pm \sqrt{(g\sin\theta)^2 - 4\left(\frac{k}{2m}\right)\left[-\left(\frac{v^2}{2} + (g\sin\theta)d\right)\right]}}{2\left(\frac{k}{2m}\right)}$$

$$x = \frac{g\sin\theta \pm \sqrt{(g\sin\theta)^2 + \left(\frac{k}{m}\right)\left[v^2 + 2(g\sin\theta)d\right]}}{k/m}$$

Because distance x must be positive,

$$x = \frac{g\sin\theta + \sqrt{(g\sin\theta)^2 + \left(\frac{k}{m}\right)\left[v^2 + 2(g\sin\theta)d\right]}}{k/m}$$

For v = 0.750 m/s, k = 500 N/m, m = 2.50 kg, θ = 20.0°, and g = 9.80 m/s^2, we have $g\sin\theta$ = (9.80 m/s^2) sin20.0° = 3.35 m/s^2 and k/m = (500 N/m)/(2.50 kg) = 200 N/m · kg. Suppressing units, we have

$$x = \frac{3.35 + \sqrt{(3.35)^2 + (200)\left[(0.750)^2 + 2(3.35)(0.300)\right]}}{200}$$

$$= \boxed{0.131\text{ m}}$$

==================FROM CHAPTER 13: For week of 23 April 2018===================

P13.30 (a) -4.77×10^9 J

(b), (c) The satellite and Earth exert forces of equal magnitude on each other, directed $\boxed{\text{downward on the satellite and upward on Earth.}}$

The magnitude of this force is 569 N

P13.33 (a) 1.84×10^9 kg/m^3

(b) 3.27×10^6 m/s^2

(c) -2.08×10^{13} J

P13.35 (a) -1.67×10^{-14} J

(b) Each particle feels a net force of attraction toward the midpoint between the other two. Each moves toward the center of the triangle with the same acceleration. They collide simultaneously at the center of the triangle.